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Malicious URL Detection Using Machine Learning

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***Abstract*— With the advent of digitalization, there has been an unparalleled increase in the use of QR codes and Uniform Resource Locators (URLs). But this increase has also been accompanied by a notable increase in malicious URLs, which presents a serious risk to cybersecurity, especially when shared via QR codes. Strong countermeasures are required because malicious QR codes carry several security risks, such as financial fraud, identity theft, and virus propagation. Traditional QR apps primarily use blacklisting methods, which are difficult to manage and don't sufficiently identify new or unknown malicious URLs. Furthermore, these approaches are static and hinder real-time threat assessment because they require users to upload QR codes for classification. The suggested system takes a novel approach to overcome these constraints. Recently, there has been an increase in the use of machine-learning techniques to enhance the detection of malicious URLs. However, for these methods to be effective, they require a large and continuously updated dataset, as they rely heavily on data. The proposed method aims at classifying URLs according to their lexical characteristics and differentiating between safe and dangerous ones. After testing the model with five different classifiers, the Extra Tree classifier outperformed the others, obtaining a 99.70\% accuracy rate. This system, in contrast to traditional approaches, uses lexical analysis to identify URL characteristics, allowing for proactive mitigation of possible threats.**

Keywords—Malicious QR Code, Machine Learning, Malicious URL, Cybersecurity.

I. Introduction

With over 2.5 billion active users, Android is the most popular operating system worldwide, accounting for 75% of the market share for mobile devices. One of Android's special threats is the Quick Response (QR) code. In 1994, the Japanese company Denso created Quick Response, the most well-known two-dimensional barcode. With the huge increase in the use of QR codes for financial transactions, contact tracing, contactless registration, user authentication, and verification in emerging markets, the number of QR code-related scams has also increased[1].

In different aspects of daily life, from product labeling to payments on the go, QR codes have become widespread and are increasingly associated with society. However, security concerns were also raised by this widespread adoption because of the potential for malicious QR codes to emerge as a threat vector. Users were exposed to the risk that their devices and user data could be irretrievably damaged by an inadvertent scan of bad codes. Efforts have been made to detect and mitigate these threats, recognizing the need to ensure the safety of users. The implementation of QR code malicious detection systems, designed to protect personal data and privacy by detecting and blocking such codes, has been a central part of this effort. These systems have played an important role in protecting themselves from cyber attacks, e.g. phishing and malware downloads, but they also mitigate the risks of data breaches by intercepting threats at their source. The system aims to reduce the risk of cyber attacks by detecting threats coming from sources, with a view to increasing confidence in QR code technology.

[17]Two recent cases illustrate the prevalence and impact of QR code fraud. In the first instance, a 30-year-old professor from the Indian Institute of Science, Bengaluru, encountered a QR code scam while selling his washing machine online. The scam resulted in the fraudulent withdrawal of Rs 63,000 from his bank account after scanning the QR code provided by the purported buyer. Similarly, another incident involved a 30-year-old homemaker in the same city who lost Rs 20,000 when attempting to sell her veena on an online marketplace. A fraudulent buyer obtained her bank account details and sent a link to her mobile phone, leading to the swift withdrawal of funds upon clicking the link. These cases underscore the urgency of addressing QR code fraud and implementing effective measures to protect users from financial losses and privacy breaches.

A URL that guides the user to the particular website or page is received after scanning a QR code. Uniform resource locator, or URL, is the name of a web address that provides the location of an online resource. It consists of many components, including the domain name (www.example.com), the protocol ("http" or "https"), and the precise path or resource (/page/index.html) that web browsers use to access and display content from the internet. URLs are required in order to explore the internet and access files, websites, and online services. But sometimes hackers hack the URL but the user is not aware of this malicious activity. Many methods can be used to solve this problem, such as the blacklist method and predefined static feature classification.

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Still, these methods do not meet the requisite standards or criteria, and therefore, fall short in terms of adequacy or effectiveness. Therefore, an effective and precise system is required to discern whether a URL is malicious or benign.

The proposed system obtains the URL and from that URL all the lexical features are obtained. These lexical features are used as input in the machine learning model which helps in identifying whether the QR code is malicious or Benign. Within the proposed system, a variety of classifiers are employed, the classifiers used are Decision Tree, Gaussian Naive Bayesian, Logistic Regression, Random Forest, and Extra Tree. Among these classifiers, the one exhibiting the highest accuracy and computational efficiency is selected for utilization. Upon the completion of model training, when a user inputs a URL, the system will predict whether the provided URL is benign or malicious.

The subsequent sections of this paper are organized as follows: Section II offers a review of the literature, Section III describes the general methodology, and Section IV shows the results, Section V concludes the research.

II. Literature Survey

This study presents an investigation into the pressing issue of malicious URLs in cybersecurity. It introduces and compares malicious URL detection machine learning(ML) models, which use different classification algorithms. The study evaluated the performance of different classification algorithms based on key confusion metrics such as True Positive Rate (TPR), False Positive Rate (FPR), recall accuracy, and F-measure, using a dataset consisting of both benign and malicious URLs.The findings indicated that the Support Vector Machine (SVM) algorithm emerged as the most effective method, achieving an accuracy of 95.4%, for discerning between benign and malicious URLs [1].

This study presented an approach to detect and classify malicious URLs using machine learning algorithms. Among the three algorithms implemented, the random forest model yielded the most accurate results. By training the random forest classifier with more balanced data, such as datasets containing nearly equal numbers of harmful and beneficial websites, the investigation demonstrated improved performance in identifying fraudulent URLs. The study emphasized the effectiveness of training a model with a database of selected attributes to anticipate new attack attempts, thereby enhancing the capability to detect fraudulent URLs. Moreover, the proposed models could be integrated into search engines or websites to provide alert messages to users, thereby mitigating the risks associated with accessing fraudulent URLs [2].

The study addressed the escalating threat posed by malicious URLs in the contemporary digital landscape, focusing on enhancing detection through supervised learning algorithms like K-Nearest Neighbor (KNN) and Logistic Regression. Results demonstrated KNN's superiority over Logistic Regression, achieving an accuracy of 90.61%. The research emphasized the significance of various URL attributes in determining malicious intent, revealing distinct characteristics conducive to encryption and deception. Through thorough visualization and analysis, the study provided valuable insights into the correlation between URL attributes and the likelihood of malicious intent, aiding in the development of robust cybersecurity measures to combat phishing attempts and safeguard sensitive data [3].

The study introduced a novel framework for phishing URL classification, relying solely on lexical-based features extracted from the URL string. It utilized various URL components like the full URL, protocol scheme, hostname, and path area, along with entropy features, suspicious words, and brand name matching using TF-IDF. Across six diverse datasets and employing eight machine learning classifiers, Random Forest consistently emerged as the top performer, achieving notably higher accuracy rates. With a concise feature set of just 30 features, the proposed framework achieved impressive accuracies of 96.25% and 94.65% on Kaggle datasets. Comparative analyses revealed that the proposed model surpassed existing approaches, achieving accuracies of 92.2%, 91.63%, 94.80%, and 96.85% on benchmark datasets, respectively [4].

This article introduces QsecR, a personal-friendly QR code scanning application designed to combat cybercriminals displaying malicious URLs. Leveraging machine learning techniques and a collection of 4,000 benign and malicious QR codes, QsecR uses a three-stage approach: URL redirect analysis, signature distribution, and malicious URL detection. The results show that QsecR has a better performance, with QsecR achieving 93.50% accuracy and 93.80% accuracy compared to Barsec's 78.63% accuracy and 77.30% accuracy. Future research will focus on improving classification and measuring the response time of secure QR code scanners to accurately identify malicious URLs [5].

This paper addresses the growing problem of QR code-related fraud in emerging markets. It proposes a solution called “Secure QR (sQR)” using deep and machine learning models. After extensive testing, the Vision Transformer model was effective in detecting malicious QR codes, achieving more than 99.96% accuracy and sensitivity. Lightweight now can integrate with existing QR systems, especially in areas where there is a lot of crime, such as India's UPI QR code. Regular training with updated information ensures that the system remains resilient to changing threats [6].

The topic of QR code security is covered in this article, with an emphasis on man-made threats including malware propagation and phishing. Examines the privacy permission of Android QR code scanner applications. These studies aim to improve the security of QR code scanners in terms of user-friendly and knowledge-based solutions. For future work, the authors plan to develop QsecR, an instant malicious URL detection model [7].

QR codes have become ubiquitous in the internet-driven society. They effectively encode data for quick access to links and information. But this accessibility also makes them a target for cyber attacks such as phishing and malware distribution. To solve this problem, the study used random forest and binary long-distance memory models to detect bad QR codes, and Bi-LSTM achieved an accuracy of 83.79%. Although accurate and reproducible, the data has some inconsistencies, indicating that URLs may be incorrectly classified as malicious [8].

This research uses URL shortening programs to combat online threats such as phishing attacks and solve privacy issues in online social networks (OSNs). This method combines a semantic vector model, URL encryption, and a backward neural network (LSTM) with k-means clustering for classification. The results showed good accuracy; It achieved 98.3% for LSTM and 99.7% for k-means clustering. This

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research presents a new technique for token-level injection and URL encryption, laying a solid foundation for on-the-fly URL detection and possible future improvements using deep learning models [9].

This paper discusses security issues with QR codes in areas such as education, fish farming, land management, and healthcare. It focuses on detecting malicious URLs in barcodes, a method used in cybercrime attacks. The authors show BarAI, an AI model using multi-classification, that the J48 decision tree achieves an accuracy of up to 90.243%. This tool shows promise in improving agricultural ecosystems' sustainability through QR codes for sustainable development [10].

This study presents an approach to detecting malicious URLs, focusing on phishing, malware, and drive-by-download websites. Utilizing the Classification based on Association (CBA) algorithm, the researchers extracted URL and webpage content-based features to effectively distinguish between benign and malicious URLs. Through experimental analysis on a real-world dataset, the CBA algorithm demonstrated comparable performance against benchmark classification models. The CBA model achieved an accuracy rate of 95.83% [11].

This article discusses the increase in cyber threats, especially among online businesses, during the Covid-19 pandemic. It uses Kaggle datasets to test various machine-learning methods to detect malicious URLs. Random Forest is the best with an F1 score and an accuracy rate of almost 99.6%. To improve its performance, future efforts could focus on training the classifier on more balanced data that balances between malicious and malicious websites [12].

This study presents a novel method aimed at preventing real-time URL spamming through the utilization of a machine-learning algorithm based on logistic regression. The proposed system incorporates a frontend plugin designed to categorize URLs into either safe or malicious categories. This approach not only effectively identifies malicious URLs but also empowers users by furnishing them with insights into the types of URLs they should avoid to safeguard themselves against cyber attacks [13].

This research addresses the major threat from cybercriminals using malicious URLs on Internet services. Heuristic research using features extracted from data and machine learning algorithms, including support vector classifiers, naive Bayes, logistic regression, K-nearest neighbors, and random forest. The results demonstrate the effectiveness of the simple method and highlight the superiority of the k-NN classifier in terms of accuracy and efficiency. This new approach shows promise in combating cyber threats in the online space [14].

This article addresses the issue of faulty URLs in the digital space and highlights the need for improved search engines. This study adopted a signature-based and machine learning-based approach using Support Vector Machine (SVM) and Random Forest (RF) algorithms. The 100-tree RF model showed good results, achieving high accuracy and low error rates. While acknowledging inconsistent data and recommending future work, this paper could benefit from additional reasoning behind algorithm selection. Overall, this study provides the best solutions and user tools to improve network security against malicious URLs [15].

This research explores the changing nature of cyber threats, particularly phishing sites, and highlights the potential of machine learning in combating these threats. Various algorithms are discussed, including SVM, logistic regression, OGD, PA, CNN, and RNN. While the findings underscore significant progress in malicious URL detection, they also highlight the need for continued research in cybersecurity, particularly in behavioral and problem-solving areas. Overall, this research provides a better understanding of combating cyber threats [16].

On surveying the papers, the findings showcased the effectiveness of various machine learning algorithms in addressing cybersecurity challenges related to QR code and URL security. Evaluation of QR code scanner applications, including QsecR and Barsec, demonstrated significant improvements in accuracy and precision. However, some of these applications use the blacklisting method that only identifies the known malicious URL listed in the dataset. In some of the papers, only 5 to 6 lexical features are included, which is insufficient for model optimization. Also, it was identified that there were only 5000 to 25000 URLs in the dataset which is not adequate for the identification of benign and malicious QR codes.

III. METHODOLOGY

The main idea is to create a malicious URL detection system that relies on categorizing URLs based on lexical characteristics. After users input URLs, these features are extracted by the system. A machine learning model is then built to determine whether a URL is malicious or benign, one classifier is selected based on accuracy among the others. This technique aims to improve user safety online by quickly detecting potentially dangerous URLs in an effort to improve cybersecurity.

A. Dataset Description

The dataset utilized in this research was gathered from Kaggle and other sources. This information includes both benign and malicious URLs. 4,09,413 malicious URLs and 5,59,977 benign URLs make up this collection. There are 9,73,425 URLs in the dataset overall. There are labels in the dataset that indicate if a URL is malicious or not.

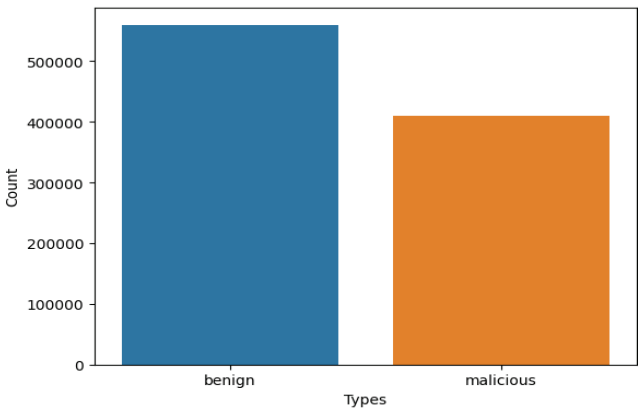

Fig. 1. Dataset distribution.

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TABLE I. CATEGORIES OF DIFFERENT URLS IN THE DATASET.

Types of Challenging URLs in dataset	Description
URL redirection	URL are redirected to the other domain.
Lengthy URL	Certain URL types that are often used in phishing attacks have longer domain names and are challenging to identify.
URL with IP address	Malicious URLs that employ IP addresses rather than domain names are more frequent targets of drive-by download attacks.
Shortened URL	Shortening services like Bitly are used to shorten the URL.

B. *Extraction of Lexical Features*

As shown in Fig3, when committing cyber crimes, cybercriminals typically use the path directory, generic top-level domain (gTLD), and second-level domain. A well-known website, like a blog, Instagram account, or TikTok, could be the domain. gTLDs may include org, net, gov, and edu. Using URL shortening services, cybercriminals try to spoof their malicious links and get around blacklists. As a result, the need to retrieve all of a URL's link-based features is growing.

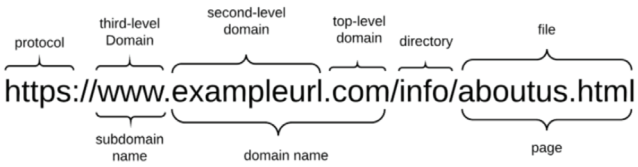

Fig. 2. Structure of URL.

The lexical features of URLs comprise a range of components that offer information about their composition and substance. These attributes generally comprise elements like the URL's length, the inclusion of particular characters or symbols (e.g. the number of subdomains, the presence of specific keywords or patterns, and the domain name itself., hyphens, underscores, and special characters). Furthermore, lexical features could include any suspicious or unusual character sequences, like random strings or sequences that mimic well-known malicious patterns. In order to facilitate classification, the system extracts the lexical features shown in Table 2 for analysis.

TABLE II. LEXICAL FEATURES.

Sr No.	Feature Name	Type
1	Length of URL	Numeric
2	URL containing IP address	Numeric
3	Count of “http”	Binary
4	Count of percent	Numeric
5	Count of questions “?”	Numeric
6	Count of hyphen “-”	Numeric
7	Count of equal “=”	Numeric
8	Length of hostname	Numeric
9	Count of “https”	Binary
10	Count of www	Binary
11	Count of dot “.”	Numeric
12	Count of atrate “@”	Numeric
13	Number of directories	Numeric
14	Number of embedded URL	Numeric
15	Length of First directory	Numeric
16	Count of digits	Numeric
17	Count of letters	Numeric
18	Use of shortening URL service	Numeric

C. *Machine Learning Classifiers*

The dataset was first read from a CSV file, and the features of each data entry were examined. Eighty percent of the dataset was used for testing, and the remaining twenty percent was used for training. To improve classifier accuracy, a feature selection process was carried out, wherein less correlated features were excluded. After that, the dataset was run through a number of machine-learning algorithms to analyze its performance. The machine learning algorithms that were used were:

- Decision Tree (DT)
- Gaussian Naive Bayesian (GNB)
- Logistic Regression (LR)
- Random Forest (RF)
- Extra Trees (ET).

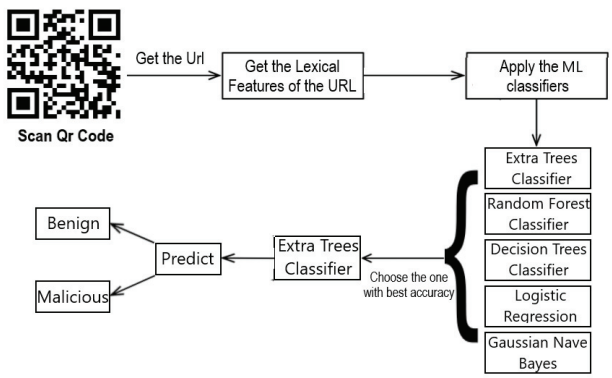

Fig. 3. Proposed System.

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II. RESULTS

A. Evaluation Metrics

In this part, the performance of machine learning (ML) algorithms on the given dataset was evaluated. The algorithms were applied to the designated test dataset, which constituted 20% of the entire dataset, in order to conduct this evaluation. To assess the performance, confusion matrices were generated, which provided metrics such as True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). These metrics are essential for calculating the accuracy of each algorithm.

- **TP (True Positive)**: Instances correctly identified as positive.
- **TN (True Negative)**: Instances correctly identified as negative.
- **FP (False Positive)**: Instances incorrectly identified as positive.
- **FN (False Negative)**: Instances incorrectly identified as negative.

TABLE III. CONFUSION MATRIX.

Actual Class		True Class	
		Malicious	Benign
Tested Class	Malicious	True Positive	False Positive
	Benign	False Negative	True Negatives

B. Performance Evaluation

Comparison of accuracies between the above-mentioned classifiers are:

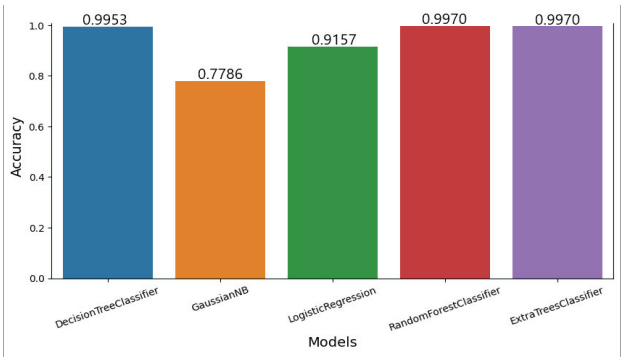

Fig. 4. Comparative Study.

For selecting an appropriate classifier, the dataset was tested and trained with a total of five different classifiers. Out of which three classifiers exhibited extraordinary accuracy, but Extra Trees was chosen because the computational time for Extra Trees (112.78648638725281) was less than that of Random Forest (136.78264236450195).

```
Test Accuracy :  99.74%
              Classification_report
              precision    recall  f1-score   support

           0       1.00      1.00      1.00    111917
           1       1.00      1.00      1.00     81961

    accuracy                           1.00    193878
   macro avg       1.00      1.00      1.00    193878
weighted avg       1.00      1.00      1.00    193878
```

Fig. 5. Extra Trees Classification Report.

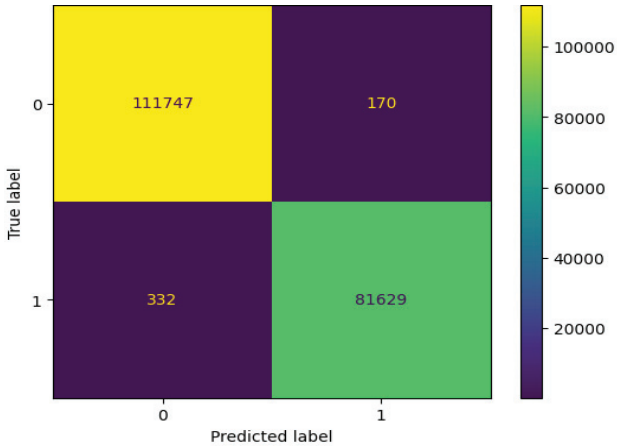

Fig. 6. Extra Trees Confusion Matrix.

In the Confusion matrix for Extra Trees Classifier out of 111,917 Benign URLs 111,747 were predicted as benign, and out of 81,961 malicious URLs 81,629 were predicted as malicious.

```
https://www.google.com/
BENIGN
https://secure.unconfirmaccounts.ga/?facebook
MALICIOUS
https://www.hotstar.com/in/home
BENIGN
 https://paypal.de-vorgangs-nummer-p361.info/51
MALICIOUS
```

Fig. 7. Prediction Results.

In Fig8, Extra Trees is utilized as the classifier for predicting whether URLs are malicious or benign. The testing conducted on various URLs determines their malicious or benign nature.

IV. CONCLUSION AND FUTURE WORK

In this paper, a machine learning model is proposed where Random Forest and Extra Trees models appear to be more suitable for the classification task due to their relatively consistent and balanced performance on both training and testing data. The Decision Tree model, while still reasonable, shows a slightly higher degree of overfitting. However, the Gaussian Naive Bayesian model appears to overfit the training data significantly, which may require further tuning or regularization to improve its generalization ability. The surge in malicious Uniform Resource Locators (URLs) transmitted via Quick Response (QR) codes poses a significant cybersecurity threat. QR codes' widespread use has raised concerns about potential misuse, leading to risks like identity theft and financial fraud. Current QR link applications often rely on inadequate blacklisting techniques

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to identify malicious URLs. A proposed solution leverages machine learning to differentiate between benign and malicious URLs, using a dataset with diverse examples of both. The model extracts 18 lexical features crucial for classification. Five algorithms were compared, with the Extra Trees Classifier proving the most accurate in predicting URL nature. This approach provides an effective countermeasure to the identified problem, enhancing URL security.

Further research endeavors to improve the accuracy of malicious URL detection by adding more factors according to their importance. Increasing the quantity of various datasets will also help with feature classification and yield a more accurate result for identifying hazardous URLs. Also, an application that scans the QR code and gives a message about whether the QR code is malicious or benign.

REFERENCES

[1] Hariri, Jana, Lujain Batouq, and Ahd Al-Jarf. "Developing a Model to Detect Malicious URLs using Different Classification Algorithms." 2024 21st Learning and Technology Conference (L&T). IEEE, 2024.

[2] DR. Upendra Shetty, and Anusha Patil. "Malicious URL Detection and Classification Analysis using Machine Learning Models." 2023 International Conference on Intelligent Data Communication Technologies and Internet of Things (IDCIoT). IEEE, 2023.

[3] Mehndiratta, Mansi, et al. "Malicious URL: Analysis and Detection using Machine Learning." 2023 10th International Conference on Computing for Sustainable Global Development (INDIACom). IEEE, 2023.

[4] Jalil, Sajjad, Muhammad Usman, and Alvis Fong. "Highly accurate phishing URL detection based on machine learning." Journal of Ambient Intelligence and Humanized Computing 14.7 (2023): 9233-9251.

[5] Rafsanjani, Ahmad Sahban, et al. "QsecR: Secure QR Code Scanner According to a Novel Malicious URL Detection Framework." IEEE Access (2023).

[6] Goel Sanchit, et al. "Sqr: High Performance Secure Qr System to Filter Against Malicious Qr Codes Using Deep Learning Techniques." Available at SSRN 4253649.

[7] Rafsanjani, Ahmad Sahban, et al. "Evaluating Security and Privacy Features of Quick Response Code Scanners: A Comparative Study." Open International Journal of Informatics 10.2 (2022): 197-207.

[8] Pawar Atharva, et al. "Secure QR Code Scanner to Detect Malicious URL using Machine Learning." 2022 2nd Asian Conference on Innovation in Technology (ASIANCON). IEEE, 2022.

[9] Afzal Sara, et al. "Url Deepdetect: A deep learning approach for detecting malicious urls using semantic vector models." Journal of Network and Systems Management 29 (2021): 1-27.

[10] Mohammed S.Al-Zahrani, Heider AM Wahsheh, and Fawaz W. Alsaade. ”Secure real time artificial intelligence system against malicious QR code links.” Security and Communication Networks 2021 (2021): 1-11.

[11] Kumi Sandra, ChaeHo Lim, and Sang-Gon Lee. "Malicious url detection based on associative classification." Entropy 23.2 (2021): 182.

[12] Shantanu, Janet, B., and R. Joshua Arul Kumar. "Malicious URL detection: a comparative study." 2021 International Conference on Artificial Intelligence and Smart Systems (ICAIS). IEEE, 2021.

[13] Rayala, Rohit, et al. "Malicious URL Detection using Logistic Regression." Authorea Preprints (2023).

[14] Raja A. Saleem, R. Vinodini, and A. Kavitha. "Lexical features based on malicious URL detection using machine learning techniques." Materials Today: Proceedings 47 (2021): 163-166.

[15] Do Xuan Cho, Hoa Dinh Nguyen, and Victor Nikolaevich Tisenko. "Malicious URL detection based on machine learning." International Journal of Advanced Computer Science and Applications 11.1 (2020).

[16] Sahoo, Doyen, Chenghao Liu, and Steven CH Hoi. "Malicious URL detection using machine learning: A survey." arXiv preprint arXiv:1701.07179 (2017).

[17] IndiaToday.https://www.indiatoday.in/technology/features/story/qr-code-scam-cases-rising-in-india-what-is-happening-how-to-stay-safefeature-2426017-2023-08-24.

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